Merra2BC User manual

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1 Introduction

The Merra2BC interpolator creates initial and boundary conditions based on MERRA-2 reanalysis (Randles et al., 2017) for a WRF-Chem simulation by interpolating chemical species mixing ratios defined on the MERRA-2 grid to WRF-Chem grid. For the initial conditions, interpolated values are written to each node of the WRF-Chem grid. For the boundary conditions, only boundary nodes are affected. The workflow of the Merra2BC interpolator is shown in Fig. 1. The utility is designed to be flexible and can be easily modified to work with different WRF-Chem chemical mechanisms and aerosol modules. Merra2BC is written in Python. The utility requires additional modules that need to be installed in the Python environment: NumPy (numpy), NetCDF4 (netcdf4) interface to work with NetCDF files, and SciPy (scipy) package.

1.1 Data requirements

The required MERRA-2 reanalysis data set including aerosol and gaseous collections is publicly available online:

- `https://disc.gsfc.nasa.gov/datasets/M2I3NVAER_5.12.4/summary`
- `https://disc.gsfc.nasa.gov/datasets/M2I3NVCHM_5.12.4/summary`

Depending on the requirements, all or one of the following aerosol and gaseous collections need to be downloaded: *inst3_3d_aer_Nv* – gaseous and aerosol mass mixing ratios, ($\mathrm{kg\,kg^{-1}}$) and *inst3_3d_chm_Nv* – carbon monoxide and ozone mass mixing ratios, ($\mathrm{kg\,kg^{-1}}$). Besides downloaded mass mixing ratios, pressure thickness *DELP* and surface pressure *PS* fields also need to be downloaded. Spatial coverage of the MERRA-2 files should include the area of the simulation domain. The time span of the downloaded files should match the start and duration of the simulation. More information regarding MERRA-2 files' specification is provided in (Bosilovich et al., 2016).

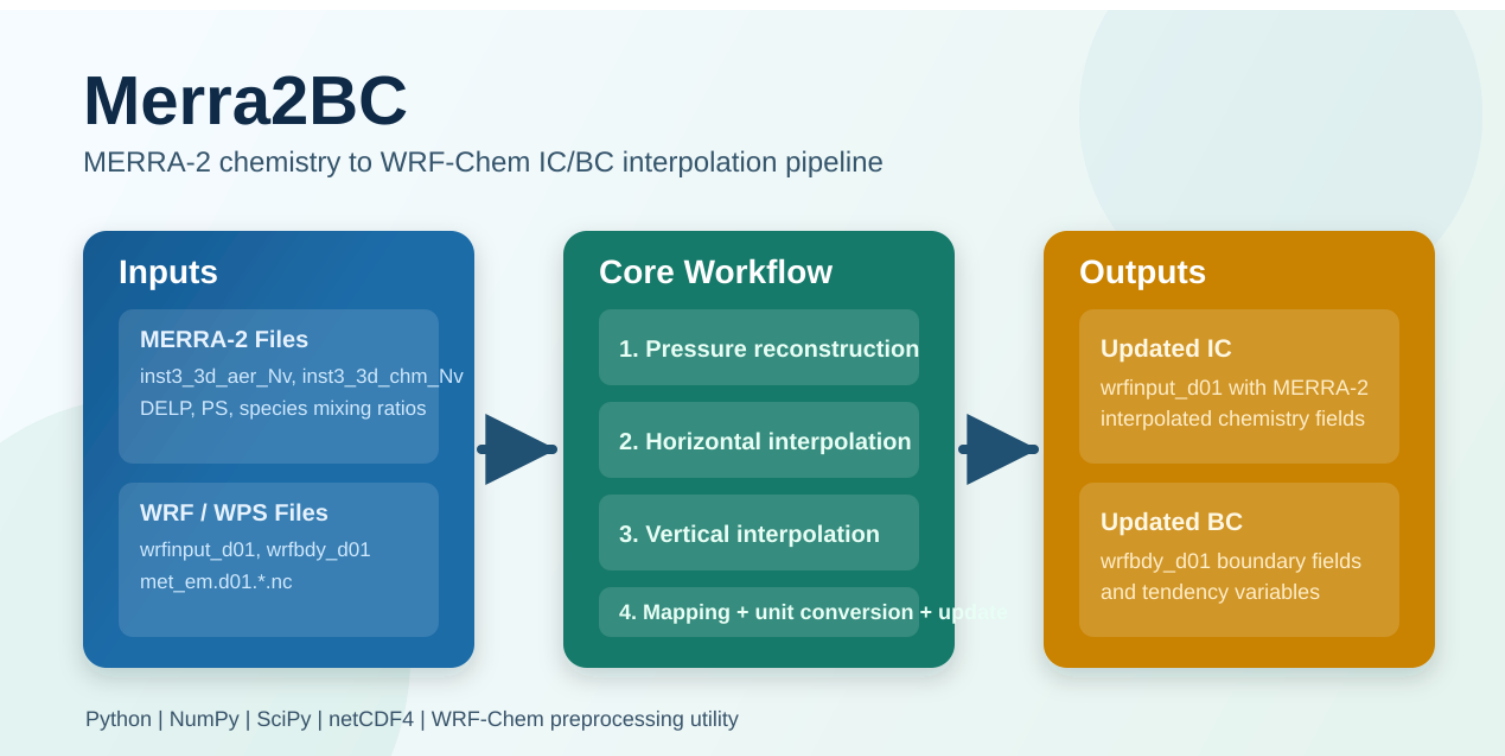

Figure 1: Merra2BC workflow overview from MERRA-2 inputs to updated WRF-Chem initial and boundary-condition files.

1.2 Reconstruction of the pressure in MERRA-2 and in WRF-Chem

Atmospheric pressure is used as a vertical coordinate. Latitude and longitude serve as the horizontal coordinates. The MERRA-2 vertical grid has 72 model layers which are on a terrain-following hybrid $\sigma - p$ coordinate. The pressure at the model top is a fixed constant, $P_{TOP} = 0.01\,\mathrm{hPa}$. Pressure at the model edges is computed by summing the *DELP* starting at P_{TOP}. A representative pressure for the layer can then be obtained by averaging pressure values on adjacent edges. Indexing for the vertical coordinate is from top to bottom; i.e., the first layer is the top layer of the atmosphere (P_{TOP}), while the 72nd layer is adjacent to the Earth's surface.

In WRF-Chem, the pressure field is not given in *wrfinput_d01* and *wrfbdy_d01* files. Hence, the pressure field must be restored using surface pressure P_{SFC} taken from *met_em_...** files created by *metgrid.exe* during the preprocessing stage. Pressure at the top of the model *wrf_p_top* and η values on half levels (*znu*) are taken from the *wrfinput_d01* file. The procedure of reconstructing the pressure from *met_em_...** files using the Python code is demonstrated in Fig. 2.

```
def get_pressure_from_metfile(metfile):
    PSFC=metfile.variables['PSFC'][:]
    WRF_Pres = np.zeros([nz,ny,nx])
    for z_level in reversed(range(nz)):
        WRF_Pres[nz-1-z_level,:]=PSFC*znu[0,z_level] +
                        (1.0 - znu[0,z_level])*wrf_p_top
    return WRF_Pres
```

Figure 2: A Python script, which reconstructs the pressure using the *met_em_...** files. nx, ny, and nz indicate the number of grid nodes in WRF-Chem domain.

1.3 Mapping chemical species between MERRA-2 and WRF-Chem

Merra2BC file *src/config.py* contains multiplication factors to convert MERRA-2 mass mixing ratios of gases given in $\mathrm{kg\,kg^{-1}}$ into ppmv. Aerosols are converted from $\mathrm{kg\,kg^{-1}}$ to $\mu\mathrm{g\,kg^{-1}}$. When using the GOCART aerosol module in WRF-Chem simulation, all MERRA-2 aerosols and gases are matched with those from WRF-Chem. We simply multiply by a factor of 10^9 to convert MERRA-2 aerosol mixing ratios given in $\mathrm{kg\,kg^{-1}}$ into $\mu\mathrm{g\,kg^{-1}}$. In the case of gases, we need to multiply MERRA-2 mass mixing ratios by a ratio of molar masses M_{air}/M_{gas} multiplied by 10^6 to convert $\mathrm{kg\,kg^{-1}}$ into ppmv, where M_{gas} and M_{air} are molar masses ($\mathrm{g\,mol^{-1}}$) of the required gas and air ($28.97\,\mathrm{g\,mol^{-1}}$), respectively. If another aerosol module is chosen in WRF-Chem, then different multiplication factors should be used.

1.4 Interpolation procedure

The interpolation procedure is based on the bivariate spline approximation for horizontal interpolation and linear interpolation for vertical interpolation. A brief description of the interpolation procedure applied to the initial conditions is presented in Alg. 1. For boundary conditions, the procedure is similar, except that additional updates of domain boundary tendencies are required and interpolation is performed for each step, where boundary conditions are applied.

1.5 Typical workflow

Here are the steps describing how to work with the Merra2BC interpolator:

1. Run *real.exe*, which will produce initial *wrfinput_d01* and boundary conditions *wrfbdy_d01* files required by the WRF-Chem simulation.

2. Download required MERRA-2 files from `https://disc.gsfc.nasa.gov/datasets/M2I3NVAER_5.12.4/summary` and `https://disc.gsfc.nasa.gov/datasets/M2I3NVCHM_5.12.4/summary`;

3. Download Merra2BC from `https://github.com/saneku/Merra2BC`.

4. Set defaults in *src/config.py* file, which contains

   a. mapping of chemical species and aerosols between MERRA-2 and WRF-Chem;

   b. paths to *wrfinput_d01*, *wrfbdy_d01*, and *met_em_...** files;

   c. a path to the downloaded MERRA-2 files.

5. Runtime arguments can override configuration values without editing *src/config.py*: `--wrf_dir`, `--wrf_input_file`, `--wrf_bdy_file` `--wrf_met_dir`, `--wrf_met_files`, `--mera_dir`, `--mera_files`, `--do_IC=true|false`, and `--do_BC=true|false`. Use "python *main.py* –help" to print all options.

Algorithm 1 Interpolation procedure applied to initial conditions.

Pressure reconstruction at each node of the MERRA-2 and WRF-Chem grids.

for each 72 vertical layers in MERRA-2 grid **do**
    Horizontal interpolation of MERRA-2 pressure on WRF-Chem latitude, longitude nodes using bivariate spline approximation (method *RectBivariateSpline* from Scipy module).
Result: MERRA-2 pressure is calculated on 72 levels but on latitude, longitude nodes of the WRF-Chem grid.

for each chemical species mixing ratio **do**
    for each 72 vertical layers in MERRA-2 grid **do**
        Horizontal interpolation of MERRA-2 species mixing ratio on WRF-Chem latitude, longitude nodes using bivariate spline approximation (method *RectBivariateSpline* from Scipy module).
    Result: MERRA-2 species mixing ratio is calculated on 72 levels but on latitude, longitude nodes of WRF-Chem grid.

    for each lat, long node of the WRF-Chem grid **do**
        Vertical linear interpolation of MERRA-2 species mixing ratio on WRF-Chem vertical coordinate (method *interp1d* from from Scipy module).
    Result: MERRA-2 species mixing ratio is interpolated at each node of WRF-Chem grid.

    Multiplying interpolated species mixing ratio by corresponding factor to convert $\mathrm{kg\,kg^{-1}}$ into ppmv or $\mu\mathrm{g\,kg^{-1}}$, depending whether it gas or aerosol.
    Updating corresponding fields in WRF-Chem $wrfinput_d01$ file by interpolated values.
Result: WRF-Chem grid is updated by interpolated values from MERRA-2 grid.

6. *real.exe* sets default boundary and initial conditions for some chemical species. Merra2BC adds interpolated values to the existing values, which may cause incorrect concentration values. To avoid this, run "python *zero_fields.py* –do_IC=true –do_BC=true", which will zero the required fields.

7. Run "python *main.py*", which will do the interpolation. As a result, files *wrfinput_d01* and *wrfbdy_d01* will be updated by interpolated MERRA-2 values. Example: "python *main.py* –wrf_dir /path/to/wrf/run –wrf_met_dir /path/to/wps/run –mera_dir /path/to/merra –do_IC=true –do_BC=true".

8. Modify the WRF-Chem *namelist.input* file at section *&chem*: set *have_bcs_chem = .true.* to activate updated boundary conditions and, if needed, *chem_in_opt = 1* to activate updated initial conditions.

9. Run *wrf.exe*.

References

Randles, C. A., da Silva, A. M., Buchard, V., Colarco, P. R., Darmenov, A., Govindaraju, R., Smirnov, A., Holben, B., Ferrare, R., Hair, J., and others. (2017). The MERRA-2 aerosol reanalysis, 1980 onward. Part I: System description and data assimilation evaluation. *Journal of Climate*, 30(17), 6823–6850.

Bosilovich, M. G., Lucchesi, R., and Suarez, M. (2016). MERRA-2: File specification GMAO Office Note No. 9 (Version 1.1).